



User Guide for Zebra Gen2X iOS app

Version: 1.0.6

May 2026

Table of Contents

<i>Introduction</i>	3
<i>Connect</i>	3
<i>Inventory</i>	4
<i>Tag Protection (Protect/Unprotect)</i>	9
<i>Tag Visibility (Enable/Disable)</i>	10
<i>Tag Focus</i>	11
<i>Tag Quieting (Quiet/Un-Quiet)</i>	12
<i>Fast ID</i>	13
<i>Inventory Protocol</i>	14
<i>Singulation</i>	15
<i>Prefilter</i>	16
<i>Custom Tag Quiet</i>	17

Introduction

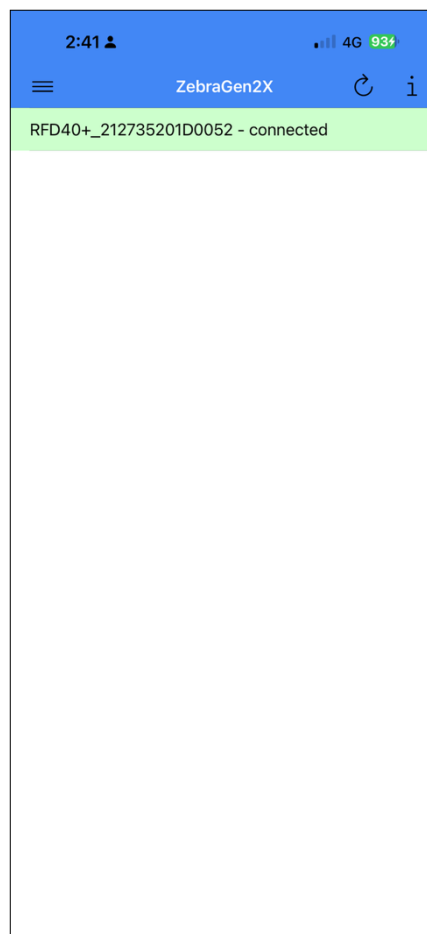
This application demonstrates the features of Impinj RFID tags. The app currently includes six major features:

- Tag Protection
- Tag Visibility
- Tag Focus
- Tag Quieting
- Fast ID
- Inventory Protocol

Connect

Connect displays all the available readers that are ready to connect. Click on the reader's name to connect to the reader.

Click Refresh to refresh the list of available readers that are ready to connect.



Inventory

The Inventory screen can be used to read tags and compare behaviors/performance using different configurations and features. This screen can be used both for standard Gen2v2 tag inventory as well as demonstrating and testing Gen2X features.

This section describes general features of the screen – please refer to specific feature sections for detailed instructions for each feature.

This screen has 4 areas:

1. Gen2X Feature Options – enable/disable options, Quiet timer
2. Inventory Tag Display - display read tags including count, RSSI and read/sec
3. Inventory Summary Bar - inventory stop timer and counts by category
4. Unique Tags Bar (including Start/Stop and Clear) – timer, unique tag count and tags/10s

The screenshot shows the Inventory screen interface. At the top, there's a status bar with the time 12:00 and battery level 76%. Below the status bar, there's a navigation bar with a back arrow and an information icon. The main content area is divided into four sections, each highlighted with a red box and a numbered callout:

- 1. Gen2X Feature Options:** This section contains several toggle switches and buttons. It includes a 'Quiet Timer' button, a 'QUIET' button, an 'Un- Quiet' button, a 'Fast ID' button, and a 'Focus' button.
- 2. Inventory Tag Display:** This section displays a table of read tags. The table has four columns: 'EPC TAG ID', 'Count', 'RSSI', and 'Read / Sec'. The rows are alternating green and grey. The first row is green and shows 'E2806810000003955FF5439' with a count of 9, RSSI of -72, and Read / Sec of 3.23. The second row is grey and shows 'E2806810000003955FF5512' with a count of 4, RSSI of -74, and Read / Sec of 2.26. The third row is green and shows 'E28011B0A506007D3DD13832' with a count of 9, RSSI of -68, and Read / Sec of 3.23. The fourth row is grey and shows 'E28068940000500650721564' with a count of 9, RSSI of -71, and Read / Sec of 3.32. The fifth row is green and shows 'E28011B0A506007D3DD13833' with a count of 6, RSSI of -70, and Read / Sec of 2.64. The sixth row is grey and shows 'E2004212A08060110180C354' with a count of 1, RSSI of -71, and Read / Sec of 0.00. The seventh row is green and shows 'E28011B0A506007D3DD15D3F' with a count of 2, RSSI of -75, and Read / Sec of 1.12. The eighth row is grey and shows 'E28011B0A506007D3DD1383B' with a count of 9, RSSI of -74, and Read / Sec of 3.50. The ninth row is green and shows 'E28011B0A506007D3DD15936' with a count of 8, RSSI of -26, and Read / Sec of 3.11. The tenth row is grey and shows 'E28011B0A506007D3DD15935' with a count of 8, RSSI of -30, and Read / Sec of 3.11. The eleventh row is green and shows 'E28011B0A506007D3DD15932' with a count of 8, RSSI of -45, and Read / Sec of 2.97. The twelfth row is grey and shows 'E28011B0A506007D3DD15934' with a count of 8, RSSI of -40, and Read / Sec of 3.11.
- 3. Inventory Summary Bar:** This section contains a 'Inventory stop timer' button and a 'Unique tags : 32' label.
- 4. Unique Tags Bar:** This section contains a 'Timer 00:02' label, a 'New Unique tags every 10s : 0' label, and a trash icon.

1. Gen2X Feature Options

Focus – enables Tag Focus feature

Quiet timer – timer for which the quiet filter is applied (default 4s)

Quiet button – quiets selected tags

Un-quiet – select to un-quiet inventoried tags

The screenshot shows the Gen2X mobile application interface. At the top, there is a status bar with the time 12:00, signal strength, Wi-Fi, and battery level at 76%. Below the status bar is a navigation bar with a back arrow, a share icon, and an information icon. The main content area is divided into two sections. The top section contains a red-bordered box with the following controls: a 'Quiet Timer' button, a 'QUIET' button, an 'Un-Quiet' button, a 'Fast ID' checkbox, and a 'Focus' checkbox. Below this box is a table with the following columns: 'EPC TAG ID', 'Count', 'RSSI', and 'Read / Sec'. The table contains 15 rows of data, with alternating green and grey background colors for each row. At the bottom of the screen, there is a summary section with the text 'Unique tags : 32' and 'Timer 00:02'. To the right of this text is a blue square icon with a white 'X' inside. Further right, there is a status bar with the text 'Y: 0 G: 23 Grey: 9 W: 0' and a 'New Unique tags every 10s : 0' label next to a trash can icon.

EPC TAG ID	Count	RSSI	Read / Sec
E28068100000003955FF5439	9	-72	3.23
E28068100000003955FF5512	4	-74	2.26
E28011B0A506007D3DD13832	9	-68	3.23
E28068940000500650721564	9	-71	3.32
E28011B0A506007D3DD13833	6	-70	2.64
E2004212A08060110180C354	1	-71	0.00
E28011B0A506007D3DD15D3F	2	-75	1.12
E28011B0A506007D3DD1383B	9	-74	3.50
E28011B0A506007D3DD15936	8	-26	3.11
E28011B0A506007D3DD15935	8	-30	3.11
E28011B0A506007D3DD15932	8	-45	2.97
E28011B0A506007D3DD15934	8	-40	3.11
E28011B0A506007D3DD15930	8	-40	3.02

Unique tags : 32
Timer 00:02

Y: 0 G: 23 Grey: 9 W: 0
New Unique tags every 10s : 0

2. Inventory Tag Display



12:00

Quiet Timer

QUIET

Un- Quiet

Fast ID

Focus

EPC TAG ID	Count	RSSI	Read / Sec
E2806810000003955FF5439	9	-72	3.23
E2806810000003955FF5512	4	-74	2.26
E28011B0A506007D3DD13832	9	-68	3.23
E28068940000500650721564	9	-71	3.32
E28011B0A506007D3DD13833	6	-70	2.64
E2004212A08060110180C354	1	-71	0.00
E28011B0A506007D3DD15D3F	2	-75	1.12
E28011B0A506007D3DD1383B	9	-74	3.50
E28011B0A506007D3DD15936	8	-26	3.11
E28011B0A506007D3DD15935	8	-30	3.11
E28011B0A506007D3DD15932	8	-45	2.97
E28011B0A506007D3DD15934	8	-40	3.11
E28011B0A506007D3DD15930	8	-40	3.03

Inventory stop timer

Unique tags : 32

Timer 00:02

New Unique tags every 10s : 0

Y: 0 G: 23 Grey: 9 W: 0

2

EPC Tag ID - EPC value read from tag

Count - Number of times read in this inventory cycle

RSSI - Received Signal Strength Indicator

Read/sec – reads per second in this inventory cycle

Colors:

- ☐ Green: Tags with a read/sec greater than the average
- ☐ Grey: Tags with a read/sec less than the average
- ☐ Yellow: Tags which were quieted
- ☐ White: Non-quieted tags which are not participating in the current round

3. Inventory Summary Bar

Inventory Stop Timer – Set a time limit to automatically stop inventory after the specified duration

Tag Counts by Category

- **Y:** Yellow
- **G:** Green
- **GREY:** Grey
- **W:** White

12:00

<

i

Quiet Timer

QUIET

☐ Un- Quiet

☐ Fast ID

☐ Focus

☐ EPC TAG ID

Count

RSSI

Read / Sec

☐ E2806810000003955FF5439

9

-72

3.23

☐ E2806810000003955FF5512

4

-74

2.26

☐ E28011B0A506007D3DD13832

9

-68

3.23

☐ E28068940000500650721564

9

-71

3.32

☐ E28011B0A506007D3DD13833

6

-70

2.64

☐ E2004212A08060110180C354

1

-71

0.00

☐ E28011B0A506007D3DD15D3F

2

-75

1.12

☐ E28011B0A506007D3DD1383B

9

-74

3.50

☐ E28011B0A506007D3DD15936

8

-26

3.11

☐ E28011B0A506007D3DD15935

8

-30

3.11

☐ E28011B0A506007D3DD15932

8

-45

2.97

☐ E28011B0A506007D3DD15934

8

-40

3.11

☐ E28011B0A506007D3DD15930

8

-40

3.02

Inventory stop timer

Unique tags : 32

Timer 00:02

Y: 0 G: 23 Grey: 9 W: 0

New Unique tags every 10s : 0

3

4. Unique Tags Bar (including Start/Stop and Clear)

12:00

76

i

Quiet Timer

QUIET

☐ Un- Quiet

☐ Fast ID

☐ Focus

☐ EPC TAG ID

Count

RSSI

Read / Sec

☐ E2806810000003955FF5439

9

-72

3.23

☐ E2806810000003955FF5512

4

-74

2.26

☐ E28011B0A506007D3DD13832

9

-68

3.23

☐ E28068940000500650721564

9

-71

3.32

☐ E28011B0A506007D3DD13833

6

-70

2.64

☐ E2004212A08060110180C354

1

-71

0.00

☐ E28011B0A506007D3DD15D3F

2

-75

1.12

☐ E28011B0A506007D3DD1383B

9

-74

3.50

☐ E28011B0A506007D3DD15936

8

-26

3.11

☐ E28011B0A506007D3DD15935

8

-30

3.11

☐ E28011B0A506007D3DD15932

8

-45

2.97

☐ E28011B0A506007D3DD15934

8

-40

3.11

☐ E28011B0A506007D3DD15930

8

-40

3.02

Inventory stop timer

Unique tags : 32
Timer 00:02

Y: 0 G: 23 Grey: 9 W: 0

New Unique tags
every 10s : 0

4

Tag Protection (Protect/Unprotect)

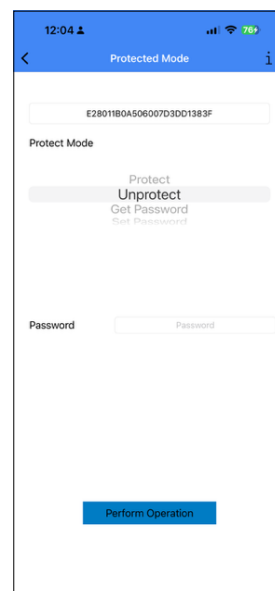
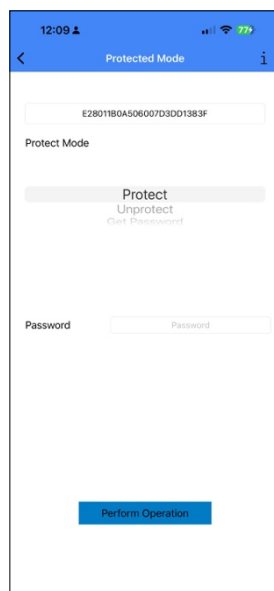
The Tag Protection feature allows you to assign a password to a specific Impinj tag, making it invisible during inventory scans.

To protect a tag:

1. Run an inventory scan.
 2. Select an Impinj tag from the inventory list.
 3. Navigate to the **Tag Protect** page.
 4. Select '**Tag Protect**' from the picker menu.
 5. Enter a password.
- **Note:** The default password for a new Impinj tag is 00000000. To change this, you must set a new password in the **Get Password & Set Password** options of the iOS app.
6. Tap the '**Perform Operation**' button.

To unprotect a tag:

1. Enter the tag's EPC in the text field.
2. Select '**Unprotect**' from the picker menu.
3. Enter the password you used to protect the tag.
4. Tap the '**Perform Operation**' button.



Tag Visibility (Enable/Disable)

This feature controls whether a protected tag is visible during an inventory scan.

To Enable Visibility:

The 'Enable Visibility' option makes a protected tag visible again during inventory.

1. Enter the EPC of the tag you want to make visible.
2. Choose '**Enable Inventory of Protected Tags**' from the picker.
3. Enter the tag's password.
4. Tap the '**Perform Operation**' button.

To Disable Visibility:

'Disable Visibility' hides the tag from inventory scans.

1. Enter the EPC of the tag you want to hide.
2. Choose '**Clear Protected Mode Config**' from the picker.
3. Enter the tag's password.
4. Tap the '**Perform Operation**' button.

The screenshot shows a mobile application interface titled 'Protected Mode'. At the top, there is a status bar with the time 12:15, signal strength, Wi-Fi, and 80% battery. Below the title bar, there is a back arrow, the title 'Protected Mode', and an information icon. The main content area displays the EPC 'E28011B0A506007D3DD1383F' in a text field. Below this, the 'Protect Mode' section contains a list of options: 'Get Password', 'Set Password', 'Enable Inventory of Protected Tags' (which is highlighted with a grey background), and 'Clear Protected Mode Config'. At the bottom of this section is a 'Password' label and a corresponding text input field. At the very bottom of the screen is a blue button labeled 'Perform Operation'.

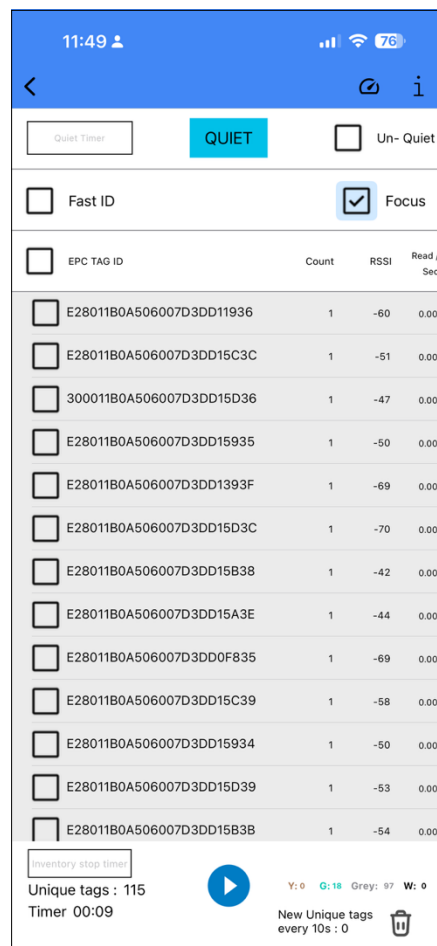
The screenshot shows the same 'Protected Mode' interface as the previous one. The EPC 'E28011B0A506007D3DD1383F' is still in the text field. In the 'Protect Mode' section, the 'Clear Protected Mode Config' option is now highlighted with a grey background, while 'Enable Inventory of Protected Tags' is no longer highlighted. The 'Password' input field and the 'Perform Operation' button remain at the bottom.

Tag Focus

The 'Tag Focus' feature ensures that Impinj tags are read only once during an inventory session.

To use this feature:

1. Launch the Impinj App and connect the reader.
2. Return to the app's Inventory page.
3. Tap on '**Focus**'.
4. Navigate to the inventory page and run the inventory. Tags will now only be read one time.
5. To deactivate this feature, uncheck '**Focus**' on the home menu.



Tag Quieting (Quiet/Un-Quiet)

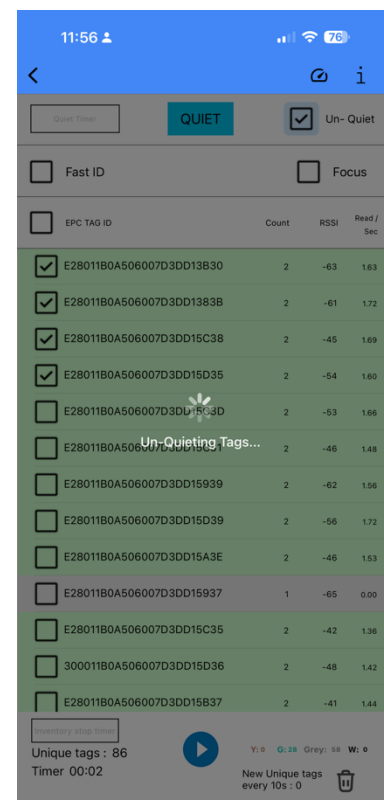
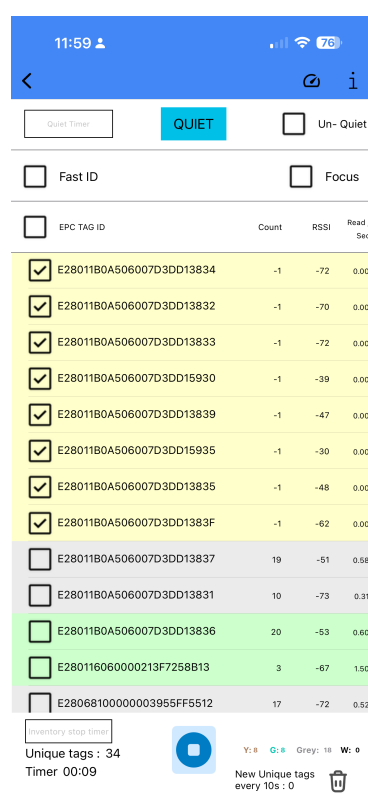
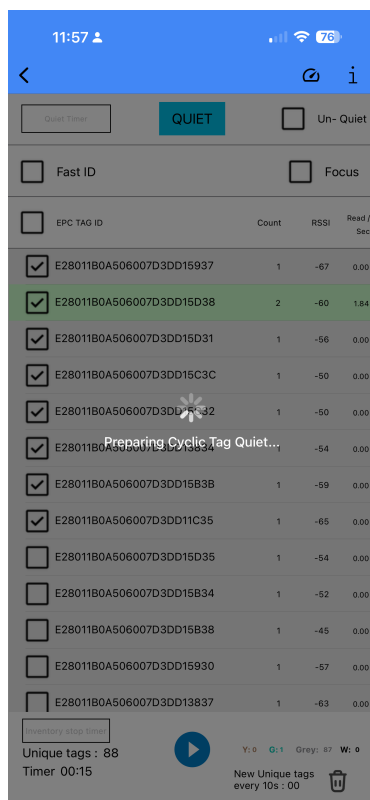
The 'Quiet' feature makes selected tags temporarily inactive, so they do not appear in the inventory.

To quiet tags:

1. Navigate to the **Inventory** page and run an inventory.
2. Select some tags from the list.
3. Press the '**Quiet**' button.
4. Run the inventory again. The selected tags will appear in yellow in the list.
5. You can continue to quiet additional tags by selecting them and pressing the 'Quiet' button again.

To make tags visible again (un-quiet):

1. Press the '**Un-Quiet**' button.
2. Run the inventory. All previously quieted tags will now be visible again.



Fast ID

Fast ID enables the RFID tags to backscatter their EPC and TID in a single response for inventory scan.

11:43

76%

<

i

Quiet Timer

QUIET

☐ Un- Quiet

☒ Fast ID

☐ Focus

☐ EPC TAG ID

Count

RSSI

Read / Sec

☐

E280116060000213F7258B13E280116020007313E4B10A7E

10

-47

1.26

☐

E28068100000003955FF5679

6

-65

0.73

☐

E280116060000213F7258B63E280116020006363E4B10A7E

10

-43

1.20

☐

E28011B0A506007D3DD15D31E28011B02000DD31EE8A03E9

7

-53

0.95

☐

E280116060000213F72579B3E2801160200061B3E4AF0A7E

10

-45

1.22

☐

E28011B0A506007D3DD15D32E28011B02000DD32EE8A03E9

8

-49

0.95

☐

E28011B0A506007D3DD15C3EE28011B02000DC3EEE8A03E9

9

-55

1.14

☐

E280116060000213F72579F3E2801160200071F3E4AF0A7E

10

-51

1.25

☐

E280116060000213F7258B53E280116020006353E4B10A7E

9

-50

1.16

☐

E28011B0A506007D3DD1383EE28011B02000D83EEE8903E9

8

-56

0.95

☐

E28068100000003955FF553A

6

-57

0.74

☐

E28011B0A506007D3DD15D3CE28011B02000DD3CEE8A03E9

7

-67

0.94

☐

E280116060000213F7258B23E280116020007323E4B10A7E

10

-43

1.21

Inventory stop timer

Unique tags : 74

Timer 00:08

Y: 0 G: 39 Grey: 35 W: 0

New Unique tags every 10s : 0

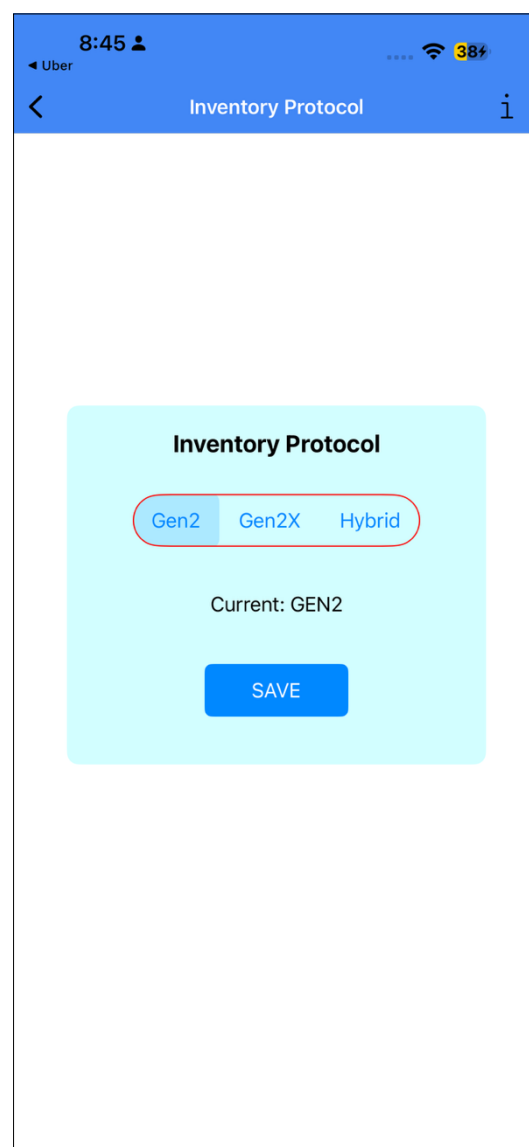
Inventory Protocol

Inventory protocol enables selection of protocol mode to determine which tags to participate in inventory.

SDK supports 3 modes for inventory protocol.

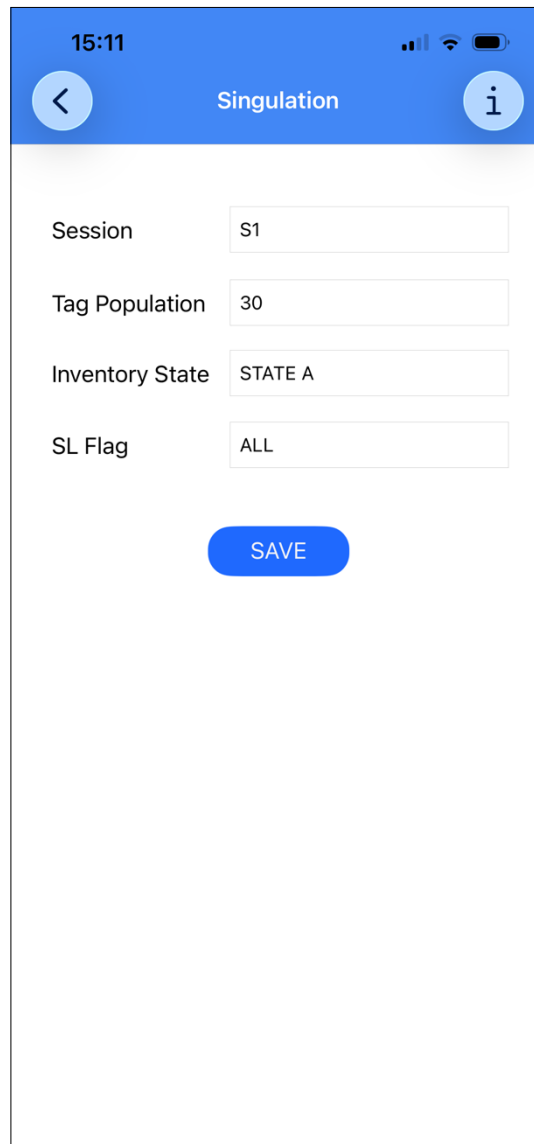
- Gen2: - Inventory all tags (Gen2 and Gen2X).
- Gen2X: - Inventory only Gen2X tags.
- Hybrid: - Inventory all tags with better read rate,

Note: FastID feature is not supported when inventory protocol is set to Gen2X mode



Singulation

User can set the required singulation in the singulation page.



The image shows a mobile application interface for the 'Singulation' settings page. At the top, there is a blue header bar with a back arrow on the left, the title 'Singulation' in the center, and an information icon on the right. The status bar at the very top shows the time as 15:11 and various system icons. Below the header, the page has a white background. It contains four labeled input fields: 'Session' with the value 'S1', 'Tag Population' with the value '30', 'Inventory State' with the value 'STATE A', and 'SL Flag' with the value 'ALL'. Each label is positioned to the left of its corresponding input field. At the bottom of the form, there is a blue rounded rectangular button with the text 'SAVE' in white capital letters.

Field	Value
Session	S1
Tag Population	30
Inventory State	STATE A
SL Flag	ALL

SAVE

Prefilter

- In prefilter user can save up to 4 prefilters.
- User cannot modify an existing prefilter. Instead of modifying, user must delete that prefilter and set a new one.
- Using delete button user can delete individual prefilter, only if the prefilters are less than or equal to four.
- Delete all button deletes all existing prefilters at once. It deletes up to 32 prefilters.
- Pointer and length are the bit numbers of the tag.

The screenshot shows a mobile application interface for setting prefilters. At the top, a blue header bar contains a back arrow, the title "Prefilter", and an information icon. Below the header, the form is organized into sections with labels and input fields: "Tag Pattern", "Pointer", "Length", "Target", "Action", and "Memory Bank". Each section has a corresponding text input field. At the bottom of the form, there is a "Pre Filter" section with a small input field containing the number "1", followed by "SAVE" and "DELETE" buttons. A large blue button labeled "DELETE ALL PREFILTERS" is positioned at the very bottom of the form area.

Custom Tag Quiet

This page allows you to set a custom quiet filter.

15:11

<

Custom Tag Quieting

i

Tag Quiet Mask 1

Tag Quiet Mask 1

Tag Quiet Mask 2

Tag Quiet Mask 2

Tag Quiet Mask 3

Tag Quiet Mask 3

Action

Action

Target

Target

SAVE TAG QUIETING FILTER

DELETE ALL PREFILTERS

15:12

<

TAG_MASK_1

i

SOB

S2B

S3B

SL_ASSERT

SOA

S2A

S3A

SL_DEASSERT